

Clinical Risk Assessment Report: Patient 45

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 14.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.67x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 4.8%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

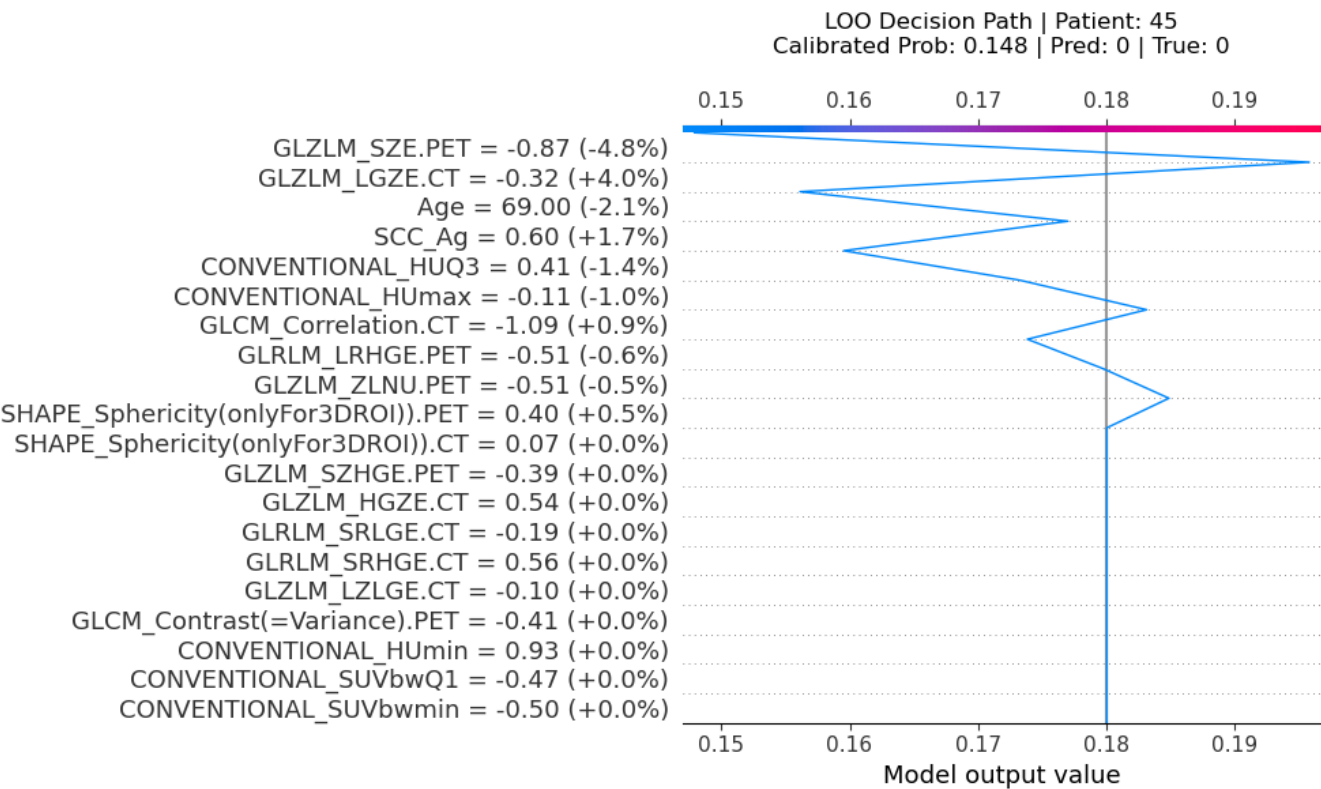
2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT)	Value: -0.32 [Bottom 20%] (+4.0%)
<i>Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.</i>	
Squamous Cell Carcinoma Antigen (SCC_Ag)	Value: 0.6 [Bottom 18%] (+1.7%)
<i>Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.</i>	
Tumor Pixel Correlation (CT) (GLCM_Correlation.CT)	Value: -1.09 [Bottom 14%] (+0.9%)
<i>Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.</i>	

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET)	Value: -0.87 [Bottom 18%] (-4.8%)
<i>Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.</i>	
Patient Age (Age)	Value: 69.0 [Avg (73th %ile)] (-2.1%)
<i>Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.</i>	
75th Percentile Tumor Density (CONVENTIONAL_HUQ3)	Value: 0.41 [Avg (55th %ile)] (-1.4%)
<i>Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.</i>	
Maximum Tumor Density (CONVENTIONAL_HUmax)	Value: -0.11 [Avg (40th %ile)] (-1.0%)
<i>Reflects the most dense solid components or calcifications within the primary tumor lesion.</i>	
Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET)	Value: -0.51 [Avg (32th %ile)] (-0.6%)
<i>Points to continuous, elongated bands of high-density or highly active tumor tissue.</i>	

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

